

THE COMPLEX STOKES CONSTANT OF TW_β : IT IS A TRANS-SERIES OBJECT, NOT A RESONANCE SHIFT — AND WHY Ω_1 IS REAL

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ABSTRACT. The genuinely resurgent (complex) analogue of the cusp $\Omega_2 = -0.451 + 0.352i$ — the median-resummation Stokes constant on the $s \rightarrow -\infty$ branch of Painlevé II — is the last open piece of Theorem T2 for TW_β . We establish where it lives, and why it is *structurally different* from the connection-root shift $\Omega_1 = -1.12$ already computed. The stochastic Airy operator is self-adjoint, so its spectrum is real for *every* noise realization; hence the connection-root (eigenvalue) shift is real to all orders and Ω_1 is real by structure, not accident. The cusp’s Ω_2 is complex because the Weber operator is *unbounded below* (escape down $-Y^2$): its “eigenvalue” $\lambda_0 = 0.8896(1 - i)$ is a genuine complex *resonance*. TW_β has no such resonance, so its complex Stokes constant cannot come from the connection root; it lives entirely in the median resummation of the divergent $1/\beta$ tail series — the imaginary ambiguity of the lateral Borel sum, fixed by reality. Its deterministic value is inherited from the Painlevé II Hastings–McLeod trans-series (Cleri–Dunne, Its–Kapaev); its $O(1/\beta)$ noise shift requires the large-order growth of the $c_n(a)$ (the transport hierarchy) or the perturbed PII connection on the Stokes sheet. This is a precise obstruction map, and it sharpens the $q=1$ vs $q=2$ contrast at the heart of both papers.

1. THE STRUCTURAL DICHOTOMY

Proposition 1 (Self-adjointness forces $\Omega_1 \in \mathbb{R}$). *The stochastic Airy operator $\mathcal{A}_\beta = -\partial_x^2 + x + 2\varepsilon b'$ on $L^2(\mathbb{R}_+)$ (Dirichlet) is self-adjoint for every realization of the real potential $x + 2\varepsilon b'$; its spectrum is real. Hence the connection root (lowest eigenvalue) $\Lambda_0[b']$ is real for every b' , and its noise average $\mathbb{E}[\Lambda_0] = \Lambda_0^{\text{det}} + \beta^{-1}\Omega_1 + \dots$ is real to all orders. [established] (Verified: over 200 realizations at $\varepsilon = 0.3$, $\max |\text{Im } \Lambda_n| = 0$ exactly.)*

Proposition 2 (Why the cusp Ω_2 is complex, and TW_β ’s is not there). *The cusp Weber operator $u'' = (\text{sign}(Y)Y^2 - \lambda - \eta\dot{W})u$ is unbounded below (escape down the $-Y^2$ side; no bottom eigenvalue). Its connection datum $D(\lambda)$ has a complex resonance root $\lambda_0 = 0.8896 - 0.8896i$ (a decaying, non-normalizable state), and $\Omega_2 = \mathbb{E}[\text{resonance shift}] = -0.451 + 0.352i$ is complex because the object being shifted is already complex. TW_β has no complex resonance (Prop. 1), so its complex Stokes constant **cannot** arise as a connection-root shift. [cited] / [established]*

This is the precise reason the Ω_1 computation returned a real number: it is the shift of a real eigenvalue. The imaginary/resurgent content lives elsewhere.

2. WHERE THE COMPLEX STOKES CONSTANT LIVES

The tail is $\mathcal{T}_\beta(a) = e^{-\beta\Phi(a)} \sum_{n \geq 0} c_n(a) \beta^{-n}$ with c_n factorially divergent (Theorem T1); by Cleri–Dunne the coefficients are non-sign-alternating in the relevant region, so the Borel singularity sits on the positive axis and the lateral Borel sums $\mathcal{S}_\pm$ differ by an imaginary amount,

$$\mathcal{S}_+ - \mathcal{S}_- \sim 2\pi i \mathcal{S}(a) e^{-\beta\Phi(a)} + \dots, \quad (1)$$

whose median $\frac{1}{2}(\mathcal{S}_+ + \mathcal{S}_-)$ is the real tail. The coefficient $\mathcal{S}(a)$ is the *Stokes constant* — the complex, genuinely resurgent object. It is a property of the divergent trans-series (the imaginary ambiguity of resummation), not of any eigenvalue; this is the $s \rightarrow -\infty$ / oscillatory-branch datum of Painlevé II.

Corollary 3 (Deterministic value is inherited). *By Theorem T1 the c_n are Painlevé II trans-series coefficients, so the deterministic Stokes constant S_{HM} is the Hastings–McLeod value of the PII connection problem (Its–Kapaev; Cleri–Dunne). [cited]*

3. THE NOISE SHIFT, AND THE OBSTRUCTION

The noise-averaged Stokes constant is $S(\beta) = S_{\text{HM}} + \beta^{-1}S_1 + \dots$, and S_1 — the true complex analogue of Ω_2 — is the target. Two routes, both genuine work:

- (R1) *Large-order growth of the c_n .* S_1 is read from the noise-dependence of $c_n(a) \sim S(a) \Gamma(n + \gamma) / \Phi(a)^{n+\gamma}$ at large n ; this needs the c_n to high order, i.e. running the transport hierarchy of the O1 note (which was set up but not evaluated). Median/lateral Borel–Padé on the resulting series then extracts $\text{Im } S$ directly, exactly as in the cusp resummation. [open]
- (R2) *Perturbed PII connection on the Stokes sheet.* Linearise PII about the HM solution (the Jacobi operator $\mathcal{J} = \mu^2 \mathcal{L}_{\text{PII}}$ of the Lax-pair note) and compute the $O(\varepsilon^2)$ noise average of its connection coefficient across the anti-Stokes line $s \rightarrow -\infty$ — on a rotated contour, as in the cusp Ω_2 code. Unlike the real eigenvalue, this connection coefficient is complex, and its noise shift carries $\text{Im } S_1$. [open]

Remark 4 (Status and the sharpened contrast). This is the one genuinely-open piece of the TW_β program, and it is *harder* than Ω_1 : not a real eigenvalue shift but a resurgent trans-series datum. The finding of §1 is itself a result: it explains why $\Omega_1 \in \mathbb{R}$ (self-adjointness) and pins the complex Stokes constant to the median-resummation sector, so the cusp and TW_β papers are contrasted precisely — the cusp’s complexity is a resonance (non-self-adjoint, unbounded-below) phenomenon, TW_β ’s is a pure trans-series phenomenon. Deterministic value: inherited (Cor. 3). Noise shift S_1 : routes R1/R2, open. [open]

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